

What are the keys to designing smaller, yet better performing systems for an IoT application?

A noteworthy point is that new hardware is being incorporated inside the chip. For instance, in surveillance cameras, the system needs to detect a person as soon as the person comes within the frame of the camera. For this application, we need motion detection algorithms running to identify the motion and sound the alarm. Conventionally, the detected video is sent to a central space where software is used for motion detection. But today designers have been able to embed chips that handle motion detection by themselves, i.e., within the chip.

Another change that has occurred is assisting the graphical user interfaces (GUIs) in systems to perform better by implementing graphics processing units within the system. NVIDIA has been historically doing that, but now multiple manufacturers have come in and are implementing this within the chip itself. Both specialised hardware working hand-in-hand with the processor, and the increase in raw performance are available to designers.

In an IoT application, there are 'always-on' devices that have to deal with a continuous trickle of data. How do these systems ensure power efficiency?

In these cases, one thing that is being done is to scale down the processing speed or the clock speed, so that they only need low power. Typically, the operating frequency in a modern mobile device is 1 GHz, but we scale it down to 20 MHz at other instances when there is enough processing power to handle the data but the power consumption remains controlled. Thus we run at a speed which is 1/50th of its standard speed, and by doing so reap the benefits of lower power consumption.

When computers became popular, we had x86 processors produced the most. For smartphones and tablets, we have ARM processors everywhere. Is there a new architecture or design coming up that would be optimal for designing systems targeting IoT?

With the increased amount of integration, it is now possible to incorporate and embed sensors inside the chips. What I believe is that more chips will come out that are even smaller than microcontrollers but support a lot of peripherals. For instance, wireless communication will be an integral part of it. There would be analogue sensors that can also make sense of the real-world conditions like humidity and temperature. These mini-SoC kind of chips can be sprinkled all over the place to provide the users with an immense amount of data collected and processed inside them.

What is the biggest challenge faced by design engineers while working with the IoT?

The biggest design challenge would be to keep up with the moving specifications. It is a fallacy that a product is frozen in specifications. Typically, a product marketing team comes up with an idea by talking with the customers and then draws up the initial requirements. Once the design process begins, the specifications will keep on changing.

A product designing firm has to have the chromosome to keep up with moving targets and specifications because ultimately the product is successful only if the customer is successful. So your ability to move with the moving target is critical.

When you build a system on the IoT, you will be required to constantly support the latest sensors into your same system. So a certain amount of modularity has to be there. If you design your architecture keeping this in mind, you will be successful. As long as you anticipate future requirements, you will be doing fine.

So how can designers ensure that their system is modular and compatible with newer products coming to the market?

Different products have different protocols and so different things apply depending on the product at hand. For instance, in the aerospace industry if you are working on in-flight entertainment systems, you need to ensure that you are in connect with the latest in the consumer electronics industry. That's because if something makes a big impact in consumer electronics, it will eventually be required by your customer for his in-flight systems as well.

What do you suggest to design engineers working on an IoT product?

First, always try to anticipate the changes and go beyond the current requirements. You should try to foresee five years down the road how things are going to evolve. If one is working on a product, it's important to foresee how it would evolve and design to achieve that.

Secondly, as a design engineer, everybody should spend more time on the actual design upfront, like the architecture. One should think about how he or she is going to design the different modules and how to provide expandability and upgradability between the modules. This is very important.

Third, a design engineer should be proud of the product designed. He should test the design himself and ensure that it works the envisioned way. A very common thing that I have seen is that design engineers just design something and then throw it over the fence to the testing team and let somebody else do the testing. Your work is your signature, so whatever output you provide should be of your quality. ●

(The complete interview can be read at electronicsforu.com)